YUXUAN ZHUANG

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RESEARCH INTERESTS

Education economics; applied AI for measurement; machine learning; causal inference; career and technical education; course tracking and placement policy; student and workforce outcomes

EDUCATION

Boston College, Lynch School of Education and Human Development 2022–present
Ph.D. Candidate in Measurement, Evaluation, Statistics, and Assessment
Dissertation: High School Pathways and Student Trajectories: Causal Evidence and New Measures

University of Texas at Austin 2018–2021
B.A. in Psychology (Honors)
Certificate in Applied Statistical Modeling | Minor in Entrepreneurship

RESEARCH IN PROGRESS

Nursing Workforce Pipelines: The Impact of High School CTE Programs
Estimates the causal effect of expanding access to high school health science programs on students' nursing credential attainment and entry into the nursing workforce.

Gatekeeping in Advanced Math: A Statewide Look at High School Tracking Policies
Documents which math courses high schools offer and what rules gate access to them, and examines whether these opportunity structures track school poverty and achievement.

Standardizing the Unstandardized: An NLP Pipeline for Private-School Transcript Research
Converts unstandardized high school transcripts into analysis-ready data, extending course-taking research to private and independent schools outside state data systems.

What Really Determines Math Placement? Evidence from a Massachusetts Public School District
Examines what drives math placement recommendations and how students' decisions to override them relate to equity and achievement.

Measuring Human Capital with Competency Embeddings
Develops a competency-based framework for measuring the skill content of education and training programs.

Student and Staff Perspectives on the Middle-to-High School Transition
Uses survey data from students and school staff to identify gaps between student expectations and high school realities during the grade 8–9 transition.

RESEARCH EXPERIENCE

Course Transcript Extraction Summers 2025 and 2026
Supervisor: Shaun Dougherty, Professor, Boston College

– Converted raw, unstandardized files that were otherwise unusable for research (transcripts, work records, and application data) into analyzable datasets.

Graduate Assistant, City Connects Research & Evaluation Team 2025–present

– Applied synthetic control methods to evaluate the causal impact of integrated student support programs on census tract-level community health outcomes.
– Analyzed principal-reported perceptions of school resource availability and program gaps across Indiana to inform statewide education policy.

TEACHING EXPERIENCE

Teaching Assistant, Data Visualization and Storytelling Summer 2026

M.S. in Data Science, Boston College

CONFERENCE PRESENTATIONS

Zhuang, Y. (2026, November). *Standardizing the unstandardized: An NLP pipeline for private-school transcript research* [Paper presentation]. Association for Public Policy Analysis and Management Fall Research Conference, Boston, MA.

Zhuang, Y. (2026, November). *What really determines math placement? Evidence from a Massachusetts public school district* [Poster presentation]. Association for Public Policy Analysis and Management Fall Research Conference, Boston, MA.

Zhuang, Y. (2026, April). *Student and staff perspectives on the challenges and opportunities of the middle-to-high school transition* [Roundtable presentation]. Annual Meeting of the American Educational Research Association, Los Angeles, CA.

Zhuang, Y. (2026, March). *Student and staff perspectives on the challenges and opportunities of the middle-to-high school transition* [Paper presentation]. Graduate Student Research Symposium, Boston College, Chestnut Hill, MA.

Zhuang, Y. (2024, April). *Exploring academic and non-academic factors that influence high school course placement decisions* [In-progress roundtable presentation]. Annual Meeting of the American Educational Research Association, Philadelphia, PA.

METHODS & TECHNICAL SKILLS

Quantitative Methods

Machine learning (text classification, transformer fine-tuning, prediction and classification), NLP for unstructured data, causal inference, quantitative data analysis, survey design and analysis

Data & Computing

Python (scikit-learn, Hugging Face Transformers, pandas, PyTorch), R, Stata | Data pipeline development (unstructured data extraction, cleaning, dataset construction) | NLP and text preprocessing | LLM fine-tuning | Data visualization

PROFESSIONAL AFFILIATIONS

American Educational Research Association (AERA)
Association for Public Policy Analysis and Management (APPAM)

RELEVANT GRADUATE COURSEWORK

Causal Design in Education Research; Economics of Education Policy; Applied AI and Machine Learning; Psychometrics I & II; Large-Scale Assessment; Assessment Programs; Multilevel Regression Models; General Linear Models; Multivariate Statistical Analysis; Intermediate Statistics